Diffused Convolutional Neural Network for Hyperspectral Image Super-Resolution

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***Abstract*— With the rapid development of deep convolutional neural networks (CNNs), super-resolution (SR) in hyperspectral image (HSI) has achieved good results. Current methods generally use 2-D convolution for feature extraction, but they cannot effectively extract spectral information. Although 3-D convolution can better characterize feature structure of HSI, it will lead to parameter redundancy, model complexity, and severe memory shortage. To address the above problems, we propose a new HSI SR method, named diffused CNN (DCNN). Specifically, spectral convolutions have been added into the enhanced convolutional neural (ECN) block, and a series of spectral convolutions are introduced in the residual network to learn features in the channel direction of different depths. Furthermore, histogram of oriented gradient (HOG) and local binary pattern (LBP) are used to retain the shape and texture information of the image, respectively, which can well represent the spatial structure of the object. To effectively make use of the extracted shallow and deep features, a feature fusion strategy is used to reinforce the reconstruction efficiency. Besides, an image enhancement module has been developed to diffuse the SR image into the image space. Extensive evaluations and comparisons show that our DCNN approach can not only recover the HSI data with richer details but also achieve superiority over several state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), feature fusion, hyperspectral image (HSI), image super-resolution (SR).**

I. Introduction

HYPERSPECTRAL image (HSI) has the unique advantage of "map-spectrum integration" [1], [2], which can accurately characterize the intrinsic structure and material properties of features and provide the possibility to distinguish and measure the composition of objects with high accuracy. Therefore, HSI is widely used in remote sensing applications such as mineral exploration [3], [4], environmental monitoring [5], climate prediction [6], and image classification [7], [8], [9], [10]. However, the long exposure time of hyperspectral systems is necessary for adequate signal-to-noise ratio (SNR), which results in low spatial resolution of HSI and limits the analysis and applications of HSI. Thus, it is meaningful to reconstruct high-resolution HSI (HR-HSI) with both spectral and spatial information from low-resolution HSI (LR-HSI).

Super-resolution (SR) is an effective method for restoring LR images to HR images, which can be roughly divided into two categories: fusion-based SR and single-image SR. Fusion-based SR is the fusion of an LR-HSI with a corresponding HR multispectral image (HR-MSI) to reconstruct an HR-HSI, which usually requires strict registration of LR-HSI and HR-MSI. Otherwise, it will cause new problems such as spectral degradation and severe artifacts. Therefore, single-image SR methods that do not require HR-MSI as auxiliary information are relatively popular in practical applications.

The existing single-image SR methods can also be divided into two classes, including traditional methods and deep learning ones. Traditional image SR are mainly low-rank tensor modeling [11], maximum a posterior (MAP) estimation [12], and sparse representation [13], [14]. Through using a prior knowledge, these methods can recover the missing information of LR images from the limited information. In particular, HSI SR reconstruction has its own characteristics, such as the spectral restoration mechanism and complexity of HSI processing. Traditional SR methods generally lack deep feature learning and semantic information extraction, which will lead to spectral degradation and structural distortion in the reconstruction results.

In recent years, deep learning has achieved excellent performance in the processing and analysis of image SR reconstruction with its end-to-end integrity training and effective implicit mapping learning [15], [16], [17], [18]. In general, convolutional neural networks (CNNs) have the ability to extract high-level fine-grained features of objects, which can effectively reduce the difficulties of manual participation and feature design. Dong et al. [19] applied CNN for image SR by designing 2-D-based hierarchical networks to learn high-level abstract features of images. Although 2-D convolution has a good ability to deal with the spatial information of HSI, it is not enough to learn the spectral information. 3-D convolution can effectively solve this problem by learning the adjacent spectral features of HSI through feature mapping

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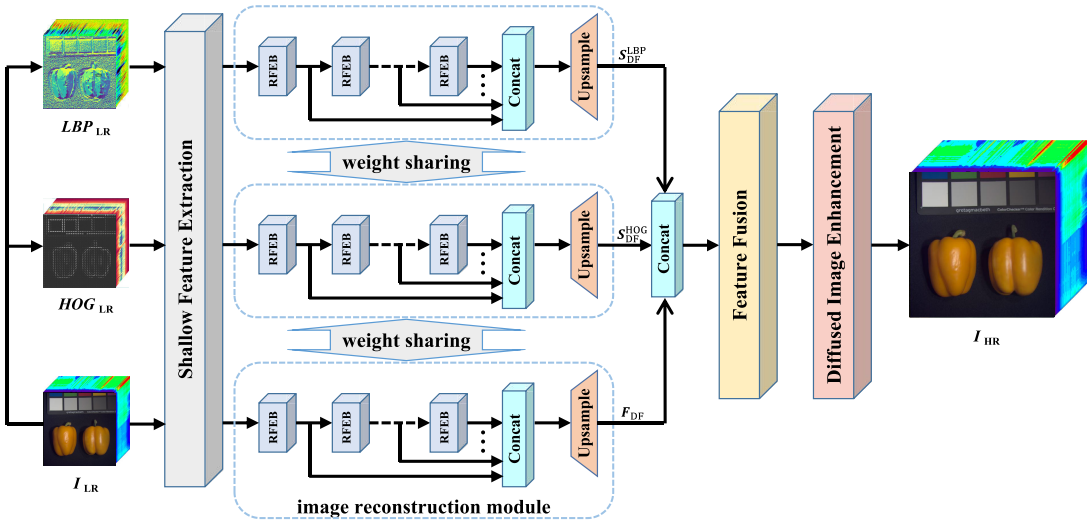


Fig. 1. Overall framework of the proposed DCNN approach.

on the channel direction of the images. Therefore, many deep learning methods based on 3-D convolution have been proposed [20], [21], but they inevitably lead to the problem of excessive number of parameters and memory overflow. Although there are many researches combining 2-D and 3-D convolution [22], [23] to reduce the number of parameters, and good results have been achieved on some public datasets, memory overflow will occur on HSI image with large spatial coverage.

Relevant studies have shown that residual networks [24] can solve the problem of network degradation through the structure of skip connections. Therefore, inspired by residual networks, this article proposed a diffused CNN (DCNN) for HSI SR. Precisely, an image enhancement module has been developed to diffuse the SR image into the image space, which fine-tunes the reconstructed image by making use of the overall information, driving the SR image closer to the real HSI, and effectively suppressing noise and artifacts. Moreover, a multilayer residual feature enhancement block (RFEB) has been carefully designed. A series of spectral convolutions are adopted to learn spectral features at different depths, which has a good performance in spectral restoration. Besides, through incorporating both kinds of shallow feature descriptors (histogram of oriented gradient (HOG) and local binary pattern (LBP) are used to retain the shape and texture information of image, respectively), the spatial structure of objects can be well-represented, and a feature fusion strategy is used to reinforce the reconstruction efficiency. Fig. 1 presents the overall framework of the proposed DCNN approach, and the contributions of this work are summarized as follows.

1) First, a diffused image enhancement module is developed, which expands the reconstructed SR images into adjacent image space and sets different weights between each diffusion image and the real image. Hence, the diffused image enhancement module is able to maximize the receptive field of the image, and the global similarity of image can be sufficiently exploited. The reconstructed image is fine-tuned by the overall information, which further suppresses noise and artifacts, driving the SR image closer to the real HSI.
2) Second, a multilayer RFEB has been carefully designed. A series of spectral convolutions are adopted to learn spectral features at different depths, which has a good performance in spectral restoration. Meanwhile, through concatenating a number of RFEB modules together, detailed textures and edge information can be well-extracted. Besides, a 1×1 convolution is adopted in the skip connection, which greatly boosts the spectral fidelity, while network complexity and memory burden are reduced as well.
3) Third, through incorporating both kinds of shallow feature descriptors (HOG and LBP are used to retain the shape and texture information of image, respectively), the spatial structure of objects can be well-represented, which plays an important role in SR reconstruction. Furthermore, to effectively make use of the extracted shallow and deep features, a feature fusion strategy is used to restore more detailed textures and sharper edges.

Extensive experiments are conducted on a wide range of HSI datasets, including two natural scenes and two remote sensing ones, and a number of state-of-the-art methods are considered in comparison. The experimental results constantly demonstrate that the proposed DCNN method achieves the best SR performance, which not only has a good effect on the shape representation of objects but also restores the internal texture information. The remainder of this article is organized as follows. Section II introduces the traditional-based and deep-learning-based HSI SR methods. A detailed description of the proposed DCNN model is provided in Section III. In Section IV, comparative experiments and ablation analysis are performed to demonstrate the effectiveness of the proposed model. Finally, conclusions are provided in Section V.

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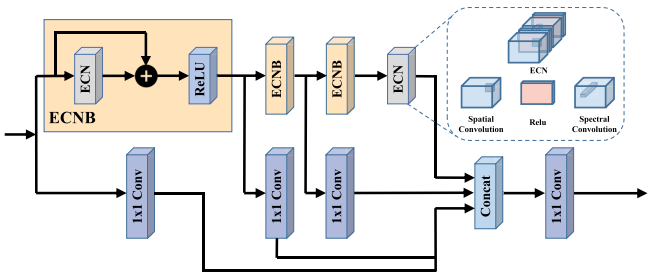


Fig. 2. RFEB of the image reconstruction module.

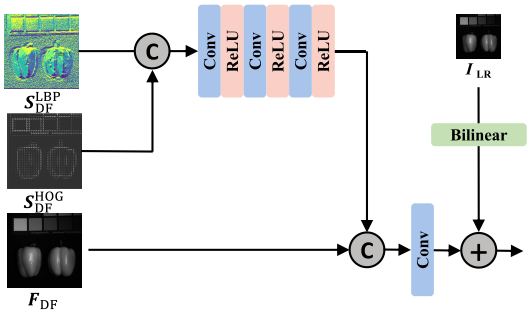


Fig. 3. Fusion architecture of the three kinds of features.

II. Related Work

A. Traditional-Method-Based HSI SR

SR reconstruction of HSI is mainly to recover detailed information from LR images. The existing traditional-method-based SR techniques are mainly based on prior knowledge of spatial texture, such as Bayesian representation [11], [25], sparse code [26], [27], [28], and wavelet transform [29], [30], [31].

Concretely, Eismann and Hardie [32] proposed a Bayesian-based fusion method to exploit the stochastic mixture of the underlying spectral scene content, but this method lacked the learning of spectral information. Subsequently, Qu et al. [33] and Sun et al. [26], respectively, introduced sparse Dirichlet distribution and gradient profile prior methods for HSI SR. Although spectral information can be better represented, they lacked robustness to spectral distortion. Dong et al. [14] proposed a joint estimation of hyperspectral dictionary and sparse code from the prior knowledge of HSI spatial–spectral sparsity. Dian et al. [34] presented a low-rank tensor modeling method to learn the correlations between spatial and spectral nonlocal similar areas. Actually, all these methods tried to compensate for the inability of considering spatial and spectral information simultaneously.

Alternatively, since HSI SR task is an ill-posed problem, many studies started to fuse high spatial MSI images to achieve better performance. For instance, Gomez et al [29] introduced wavelet transform to carry out MSI and HSI fusion. Li et al. [35] proposed a coupled sparse tensor factorization-based approach by fusing a LR-HSI with an HR-MSI. This approach factorized HSI as a sparse core tensor multiplication by dictionaries of three modes. However, due to the complexity of real environment, the expression ability of these prior knowledge is limited. Therefore, these methods do not perform well in most practical applications, especially when the image degradation is unknown [36], [37], [38], which is prone to spectral distortion.

In general, most conventional methods need a prior knowledge of degradation models. In addition, the quality of HSI reconstruction remains far below the requirements of practical applications due to the lack of end-to-end learning and inadequate use of spectral information.

B. Deep-Learning-Based HSI SR

Relying on the strong ability of convolutional operation to learn image features, many deep-learning-based HSI SR methods have been proposed in the literature [39], [40], [41], [42]. Depending on the difference in convolutional structures, these methods can be mainly divided into two categories, i.e., 2-D convolutions and 3-D ones.

With respect to 2-D convolutions, Xie et al. [43] used 2-D deep neural network with nonnegative matrix factorization strategy to extract hybrid feature matrix. Yao et al. [44] developed a two-stream 2-D convolutional autoencoder to jointly decompose MSI and HSI data into spectrally meaningful bases and corresponding coefficients. Dong et al. [45] and Tai et al. [46] improved the 2-D CNN models through incorporating a compact hourglass-shaped CNN structure and deep recursive residual network, respectively. Nevertheless, 2-D convolution mainly deals with the spatial structure of HSI and lacks the exploration of spectral information.

Fortunately, 3-D convolution can extract both spectral and spatial information of HSI. Precisely, Mei et al. [20] first introduced 3-D full convolution into the SR tasks. Yang et al. [21] proposed a multiscale wavelet 3-D CNN for HSI SR. Palsson et al. [47] first used principal component analysis to reduce the dimensionality of both MSI and HSI data, and then a 3-D CNN model is adopted to accomplish HSI SR. Although 3-D convolution can effectively extract joint spectral–spatial features, it often leads to overfitting and generates a large number of parameters. Wang et al. [48] extracted spatial and spectral features through spatial and spectral separable 3-D convolution, and Li et al. [22] and Li and Wang [23] improved the 3-D separable convolution to further reduce the computing cost, but memory overflow will also occur on HSI image with large spatial coverage.

In practice, networks using 2-D convolution can be trained much faster than 3-D convolution. Meanwhile, continuous increase in the 2-D convolution network depth can sufficiently exploit the spectral information of HSI data. In this article, 2-D CNN is mainly concerned.

III. Methods

In this section, the proposed DCNN is detailed. First, the overall DCNN framework is presented. Subsequently, the RFEB is introduced, and then the feature fusion module is given. Finally, the diffused image enhancement module is provided.

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A. DCNN Framework

As shown in Fig. 1, the DCNN model mainly consists of four modules: image reconstruction, shallow feature extraction, feature fusion, and diffused image enhancement. We denote $\boldsymbol{I}_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times C_{\mathrm{in}}}$ as the input LR-HSI and $\boldsymbol{I}_{\mathrm{HR}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C_{\mathrm{in}}}$ as the original HR-HSI, where H, W, C_{in}, and r are the height, width, input channel number, and scale factor, respectively. HOG features describe the edge and contour information of the image by calculating the gradient of pixel, which operates on the local unit (cell) of the image. HOG can not only have good invariance to the geometric and optical deformation of the image but also well describe the local shape information and is not sensitive to illumination. Alternatively, LBP features are used to reflect the texture information of the region by comparing the central pixel with adjacent pixels, which has a good performance when the background covers messy noise edges. Through taking advantage of the both kinds of feature descriptors, the spatial texture structure can be well-represented, which plays an important role in SR reconstruction. Besides, due to the simple calculation procedure of both HOG and LBP, the efficiency of the two feature extraction methods can be guaranteed. Hence, in our proposed DCNN model, HOG and LBP are used to retain the shape and texture information of image.

Specifically, DCNN extracts spatial information $\boldsymbol{S}_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times 2C_{\mathrm{in}}}$ (including HOG and LBP) from $\boldsymbol{I}_{\mathrm{LR}}$ to achieve sufficient edge and texture information. Both $\boldsymbol{I}_{\mathrm{LR}}$ and $\boldsymbol{S}_{\mathrm{LR}}$ are fed into the shallow feature extraction function $\mathcal{E}$ with 3 $\times$ 3 convolutional layer to generate shallow features $\boldsymbol{F}_0 \in \mathbb{R}^{H\times W\times C}$ and $\boldsymbol{S}_0 \in \mathbb{R}^{H\times W\times 2C}$ as

$$\boldsymbol{F}_0, \boldsymbol{S}_0 = \mathcal{E}^{\mathrm{l}}(\boldsymbol{I}_{\mathrm{LR}}),\ \ \mathcal{E}^{\mathrm{s}}(\boldsymbol{S}_{\mathrm{LR}}) \tag{1}$$

where $\mathcal{E}^{\mathrm{l}}$ and $\mathcal{E}^{\mathrm{s}}$ obtain shallow features of input image and spatial information, respectively. C is the feature channel number. In the early visual processing, mapping the low-dimensional feature space to the high-dimensional feature space is beneficial to the training and optimization of the model. In the SR task, input images contain rich information. By mapping to the high-dimensional feature space, the spatial and spectral information of the input images can be preserved and reinforced, which is an important step for the subsequent reconstruction task. To reduce memory usages and computational costs, C is empirically set as 64. Then, we extract the deep features $\boldsymbol{F}_{\mathrm{DF}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C}$ and $\boldsymbol{S}_{\mathrm{DF}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times 2C}$ from the shallow features as

$$\boldsymbol{F}_{\mathrm{DF}}, \boldsymbol{S}_{\mathrm{DF}} = \mathcal{R}_{\mathrm{REC}}(\boldsymbol{F}_0), \quad \mathcal{R}_{\mathrm{REC}}(\boldsymbol{S}_0) \tag{2}$$

where $\mathcal{R}_{\mathrm{REC}}$ is the image reconstruction module and it is made up of D RFEBs, a concatenate operation, and an upsampling operation. More specifically, intermediate features $\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \boldsymbol{F}_D$ and output deep feature $\boldsymbol{F}_{\mathrm{DF}}$ are extracted block by block as

$$\begin{aligned} \boldsymbol{F}_i &= \mathcal{R}_{\mathrm{RFEB}_i}(\boldsymbol{F}_{i-1}), \quad i = 1, 2, \ldots, D \\ \boldsymbol{F}_{\mathrm{DF}} &= \mathcal{R}_{\mathrm{UP}}(\mathcal{C}(\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \boldsymbol{F}_D)) \end{aligned} \tag{3}$$

where $\mathcal{R}_{\mathrm{RFEB}_i}$ denotes the ith RFEB and $\mathcal{R}_{\mathrm{UP}}$ is the mapping function of the upsampling layer. $\mathcal{C}$ is the concatenating operation. By sharing parameters of the image reconstruction module, $\boldsymbol{S}_{\mathrm{DF}}$ is obtained simultaneously. To effectively fuse spatial and spectral features as well as shallow and deep features, a feature fusion for synthesizing the above features is developed, which is formulated as follows:

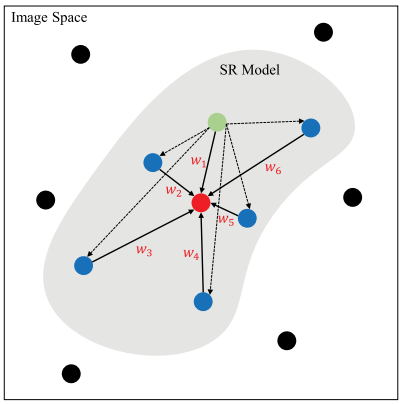


Fig. 4. Motivation of the diffused image enhancement module. Red and green points, respectively, denote the HR-HSI and SR HSI, while blue dots represent the image expanded by the green point.

$$\boldsymbol{I}_{\mathrm{REC}} = \mathcal{M}\left(\boldsymbol{F}_{\mathrm{DF}}, \boldsymbol{S}_{\mathrm{DF}}, \hat{\boldsymbol{I}}_{\mathrm{LR}}\right) \tag{4}$$

where $\boldsymbol{I}_{\mathrm{REC}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C_{\mathrm{in}}}$ is the reconstructed image and $\hat{\boldsymbol{I}}_{\mathrm{LR}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C}$ is obtained from $\boldsymbol{I}_{\mathrm{LR}}$ through bicubic upsampling. $\mathcal{M}$ is the fusion strategy, which is detailed in Section III-C. Although $\boldsymbol{I}_{\mathrm{REC}}$ contains rich texture and background information, it uses convolution operation to learn the corresponding domain information of each pixel, which deteriorates the quality of reconstruction. Hence, a diffused image enhancement module has been designed to optimize the final SR HSI, which can be described as follows:

$$\boldsymbol{I}_{\mathrm{SR}} = \mathcal{R}_{\mathrm{DIE}}(\boldsymbol{I}_{\mathrm{REC}}) \tag{5}$$

where $\boldsymbol{I}_{\mathrm{SR}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C_{\mathrm{in}}}$ represents the reconstructed SR HSI. $\mathcal{R}_{\mathrm{DIE}}$ refers to the diffused image enhancement module, which is detailed in Section III-D. Through this module, diffused images can be effectively used to learn the relationship between adjacent images, so as to improve the accuracy of reconstructed images.

B. Residual Feature Enhancement Block

Fig. 2 displays the RFEB of the image reconstruction module. To fully extract the correlations between different spectra, an enhanced convolutional neural block (ECNB) is developed. The ECN is composed of two 2-D convolutions with kernel sizes of 3 $\times$ 3 and 1 $\times$ 1, respectively, and a ReLU layer is added after each convolution to prevent gradient disappearance and overfitting. Concretely, convolution with a kernel size of 3 $\times$ 3 can extract the spatial structure, while convolution with a kernel size of 1 $\times$ 1 can well-represent the information in the spectral domain. Through sequential combination of the above two convolutions, ECN can intensively characterize the spatial and spectral information of HSI.

Although 3-D convolution is capable of extracting both spatial and spectral features of HSI, it is obvious that ECN

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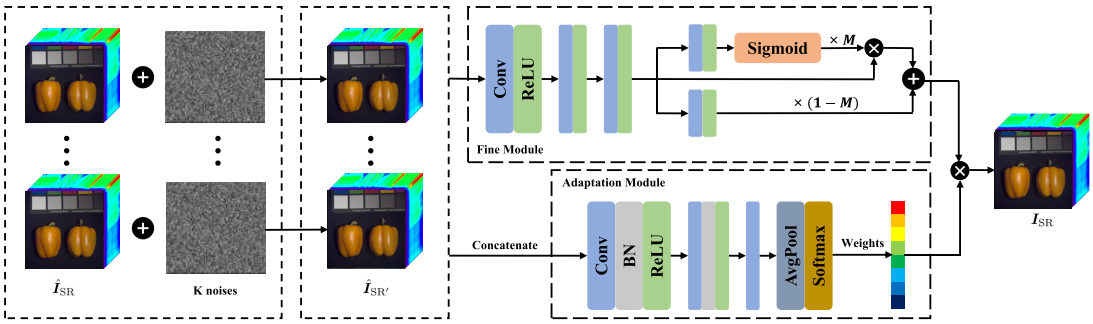


Fig. 5. Structure of the diffused image enhancement module.

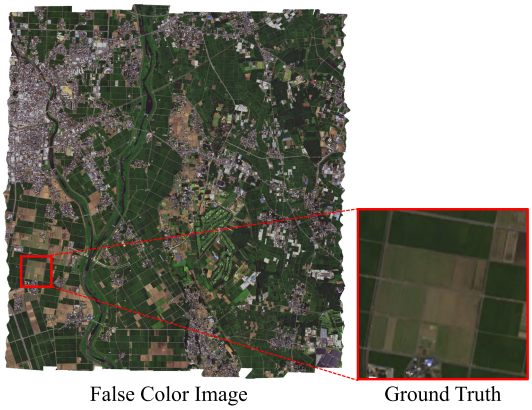


Fig. 6. Image on the left is a display of the false color image of the Chikusei dataset, and the image on the right is a display of the ground truth of the test set.

has a lower number of parameters and computational requirements. To further extract deeper information and reduce the complexity of DCNN, RFEB is established. According to (1), given the input feature maps $\boldsymbol{F}_{i,0}$ of the ith RFEB, intermediate features $\boldsymbol{F}_{i,1}$, $\boldsymbol{F}_{i,2}, \ldots, \boldsymbol{F}_{i,L}$ are acquired by L ECNBs and a spectral convolution. RFEB is calculated by the following steps.

An ECNB is formed by an ECN and the ReLU activation function, which is denoted by $\mathcal{R}_{\mathrm{ECNB}}$

$$\begin{aligned} \mathcal{R}_{\mathrm{ECNB}}\left(\boldsymbol{F}_{i,j}\right) &= \mathrm{ReLU}\left(\boldsymbol{F}_{i,j} + \mathcal{R}_{\mathrm{ECN}}\left(\boldsymbol{F}_{i,j}\right)\right) \\ \boldsymbol{F}_{i,j} &= \mathrm{Conv}_1\left(\mathcal{R}_{\mathrm{ECNB}_{i,j}}\left(\boldsymbol{F}_{i,j-1}\right)\right), \quad j = 1, 2, \ldots, L \end{aligned} \tag{6}$$

where $\mathcal{R}_{\mathrm{ECNB}_{i,j}}$ is the jth ECNB in the ith RFEB, and $\mathcal{R}_{\mathrm{ECN}}$ represents the ECN module. Conv_i represents a convolution operation with a kernel size of $i \times i$. To enhance the learning of spectrum, network depth is continuously increased and deeper spectral information can be extracted. To prevent the disappearance of gradient and network degradation, 1×1 convolutions are added to each skip connection of ECNB, which strengthens residual feature maps extracted from different depths. At the same time, the number of channels is reduced by half to reduce the burden of the network. Finally, spectral information represented by residual feature maps with different depths are not the same, which are then concatenated together and mapped into $\boldsymbol{F}_0$ space to obtain deep features.

C. Feature Fusion Module

The HOG features describe the edge and contour information of the image by calculating the gradient of pixel, while LBP features compare the central pixel with the adjacent pixels to reflect the texture information of the area. If all the three kinds of features (including the deep feature $\boldsymbol{F}_{\mathrm{DF}}$) are directly concatenated, the performance will deteriorate in the process of model training (quantitative results are provided in Section IV-D Ablation Studies). In our DCNN, they are fused in an elegant way, as displayed in Fig. 3. Specifically, both HOG and LBP are combined together, and then a number of convolution operations are applied to extract further features. After that, the achieved features are combined with the deep features. Mathematically, the fusion module is formulated as

$$\begin{aligned} \boldsymbol{S}_{\mathrm{SR}} &= \mathcal{R}_{\sigma}^{1}\left(\mathcal{R}_{\sigma}^{3}\left(\mathcal{R}_{\sigma}^{3}\left(\boldsymbol{S}_{\mathrm{DF}}\right)\right)\right) \\ \boldsymbol{I}_{\mathrm{REC}} &= \mathrm{Conv}_3(\mathcal{C}(\boldsymbol{S}_{\mathrm{SR}}, \boldsymbol{F}_{\mathrm{DF}})) + \hat{\boldsymbol{I}}_{\mathrm{LR}} \end{aligned} \tag{7}$$

where $\mathcal{R}_{\sigma}^{i}$ integrates a convolution operation with a kernel size of $i \times i$ and a ReLU activation. $\boldsymbol{S}_{\mathrm{SR}} \in \mathbb{R}^{(r \cdot H) \times (r \cdot W) \times C_{\mathrm{in}}}$ is the achieved spatial features, which is then integrated with $\boldsymbol{F}_{\mathrm{DF}}$. $\hat{\boldsymbol{I}}_{\mathrm{LR}}$ is the shallow features obtained from $\boldsymbol{I}_{\mathrm{LR}}$ through bicubic upsampling. Since there is a strong similarity between the input LR images and the output HR images, cross-space skip connection can be used to upsample the input LR images to HR space by bilinear. (Bilinear uses bilinear filtering that does not produce ringing and can also eliminate the noise to a certain extent. Due to the lower computational requirement, a bilinear interpolation method is adopted.) Through the fusion module, spatial information can be better represented to learn sharper edges and recover finer texture details. Furthermore, the integration of shallow and deep features can avoid neglecting the shallow information with the increase in network depth.

D. Diffused Image Enhancement

Fig. 4 depicts the motivation of the diffused image enhancement module. Given an image space, it contains the original HR-HSI $\boldsymbol{I}_{\mathrm{HR}}$ (red point) and the SR HSI $\hat{\boldsymbol{I}}_{\mathrm{SR}}$ (green point). It can be easily derived that $\boldsymbol{I}_{\mathrm{HR}}$ and $\hat{\boldsymbol{I}}_{\mathrm{SR}}$ are very close, but they do not coincide completely. So we get a set of SR images (blue points) by spreading out the green point. In this reconstruction set, the image that is closer or farther to $\boldsymbol{I}_{\mathrm{HR}}$ than $\hat{\boldsymbol{I}}_{\mathrm{SR}}$ exists. On this basis, we assign different weights to

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TABLE I
ABLATION STUDY OF VARIOUS MODULES IN DCNN ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

Modules	Different combinations of modules							
image reconstruction	✔	✔	✔	✔	✔	✔	✔	✔
shallow feature extraction	✘	✔	✘	✘	✘	✔	✔	✔
feature fusion	✘	✘	✔	✘	✔	✘	✔	✔
diffused image enhancement	✘	✘	✘	✔	✔	✔	✘	✔
PSNR↑	40.285	40.756	40.894	40.356	40.915	40.803	41.341	**41.702**
SSIM↑	0.977	0.980	0.980	0.977	0.980	0.980	0.982	**0.983**
SAM↓	3.398	3.362	3.283	3.391	3.251	3.379	3.151	**3.007**

TABLE II
INFLUENCE OF DIFFERENT CONVOLUTION MODELS ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

Type	PSNR↑	SSIM↑	SAM↓	params (10^5)
2DC	40.117	0.952	3.154	1.287
3DC	**41.714**	**0.983**	**2.987**	3.856
ECN	41.702	**0.983**	3.007	**0.632**

TABLE III
INFLUENCE OF THE NUMBER OF RFEB ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

Number of RFEBs	2	3	4	5	8
PSNR↑	41.04	41.30	**41.70**	41.63	41.55
SSIM↑	0.972	0.978	**0.983**	0.981	0.980
SAM↓	3.115	3.078	**3.007**	3.013	3.021
Params (10^5)	**9.18**	10.79	12.41	14.02	15.63

TABLE IV
INFLUENCE OF THE NUMBER OF ECNB ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

Number of ECNBs	2	3	4	5
PSNR↑	40.93	41.70	**41.74**	41.73
SSIM↑	0.969	0.983	**0.985**	**0.985**
SAM↓	3.123	3.007	3.013	**3.004**
Params (10^5)	**10.60**	12.41	14.42	16.65

TABLE V
COMBINATION MANNER IMPACT OF THE THREE KINDS OF FEATURES ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

combination manner	addition	concatenation	fusion
PSNR	40.761	40.803	**41.702**

this set of diffused images, which are used to improve the final result.

According to the above idea, the structure of the proposed diffused image enhancement module is detailed in Fig. 5. First, a number of perturbations (K) are added to the initially obtained image $\hat{\boldsymbol{I}}_{\mathrm{SR}}$ to obtain a set of K images. The disturbance added in this article is the Gaussian noise with low noise level, and the kth image with noise can be denoted as

$$\hat{\boldsymbol{I}}^k_{\mathrm{SR}'} = \hat{\boldsymbol{I}}_{\mathrm{SR}} + n_k, \quad k = 1, 2, \ldots, K \tag{8}$$

where $\hat{\boldsymbol{I}}^k_{\mathrm{SR}'} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C_{\mathrm{in}}}$ are a set of diffused images. Since simply adding noise deteriorates the quality of reconstructed images, each $\hat{\boldsymbol{I}}^k_{\mathrm{SR}'}$ is initially fine-tuned. The fine-tuning module $\mathcal{R}_{\mathrm{FM}}$ is defined as

$$\boldsymbol{I}_{\mathrm{REFINE}} = \mathcal{R}_{\mathrm{FM}}\left(\hat{\boldsymbol{I}}_{\mathrm{SR}'}\right) \tag{9}$$

where $\boldsymbol{I}_{\mathrm{REFINE}}$ are the images after fine-tuning, and $\hat{\boldsymbol{I}}_{\mathrm{SR}'} = \{\hat{\boldsymbol{I}}^1_{\mathrm{SR}'}, \ldots, \hat{\boldsymbol{I}}^K_{\mathrm{SR}'}, \hat{\boldsymbol{I}}_{\mathrm{SR}}\}$ are diffused images with SR image. Specifically, in the fine-tuning module, the data are grouped together to facilitate the calculation process.

After that, each $\hat{\boldsymbol{I}}_{\mathrm{SR}}$ is extended to a set of images $\hat{\boldsymbol{I}}_{\mathrm{SR}'}$. These images differ from HR-HSI in image space, so different weights should be assigned to each image. The weights $\boldsymbol{W} \in \mathbb{R}^{K+1}$ can be obtained through the adaptation module, which are computed as follows:

$$\begin{aligned} \boldsymbol{W} &= \mathrm{Softmax}\left(\mathrm{AvgP}\left(\mathrm{Conv}_3\left(\mathcal{R}_{\mathrm{CBR}}\left(\mathcal{R}_{\mathrm{CBR}}\left(\hat{\boldsymbol{I}}_{\mathrm{SR}'}\right)\right)\right)\right)\right) \\ \sum \boldsymbol{W} &= 1 \end{aligned} \tag{10}$$

where Softmax and AvgP denote the softmax activation function and the averagepooling function, respectively. $\mathcal{R}_{\mathrm{CBR}}$ is a submodule in the fine-tuning module and consists of Convolution-BatchNormals-AveragePooling, which is used to enhance the relationship between $\hat{\boldsymbol{I}}^k_{\mathrm{SR}'}$ and $\hat{\boldsymbol{I}}_{\mathrm{SR}}$. Finally, the weight is multiplied by the fine-tuning image and then summed up to obtain the final SR image

$$\boldsymbol{I}_{\mathrm{SR}} = \boldsymbol{W}_{K+1} \times \hat{\boldsymbol{I}}_{\mathrm{SR}} + \sum_{k=1}^{K}\left(\boldsymbol{W}_k \times \boldsymbol{I}^k_{\mathrm{REFINE}}\right) \tag{11}$$

where $\boldsymbol{I}_{\mathrm{SR}} \in \mathbb{R}^{(r\cdot H)\times(r\cdot W)\times C_{\mathrm{in}}}$ represents the reconstructed SR image. Through the diffused image enhancement module, the relationship between the SR image and the original HR-HSI can be explored more thoroughly, so as to better approximate the original HR-HSI and achieve high-quality SR performance.

IV. EXPERIMENTAL RESULTS

A. Dataset

Experiments are conducted on a wide range of HSI datasets, including two natural scene sets (CAVE [49] and Harvard [50])

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TABLE VI
INFLUENCE OF DIFFERENT LOSS FUNCTIONS ON THE CAVE DATASET (THE SCALE FACTOR IS 2)

Loss Function	PSNR↑	SSIM↑	SAM↓
L1	**41.702**	**0.983**	**3.007**
L2	41.324	0.981	3.186
0.5 × (L1+L2)	41.461	0.981	3.165

and two satellite remote sensing data (Chikusei and Yellow River Estuary).

1) CAVE: The CAVE dataset is gathered by cooled charge coupled device (CCD) camera and contains 32 different indoor scenes. Each image includes 31 spectral bands that are uniformly measured in the wavelength ranging from 400 to 700 nm at an interval of 10 nm. The spatial resolution of each HSI is 512 × 512 pixels.
2) Harvard: The Harvard dataset is obtained by Nuance FX, CRI INC. camera and consists of 77 HSIs of real-world indoor or outdoor scenes under daylight illumination. Each image covers 31 spectral bands ranging from 420 to 720 nm at an interval of 10 nm. The spatial dimensions of each HSI are 1024 × 1392 pixels.
3) Chikusei: The Chikusei dataset is captured by the Headwall Hyperspec-visible and near infrared (VNIR)-C imaging sensor in Chikusei, Ibaraki, Japan, on July 29, 2014. It contains 128 bands in the spectral range of 363–1018 nm, and the ground sampling distance is 2.5 m.
4) Yellow River Estuary: The Yellow River Estuary dataset is obtained from a visible shortwave infrared (SWIR) advanced HSI (AHSI) amounted on China's Gaofen5 satellite. It contains 330 bands ranging from 0.4 to 2.5 μm. About 150 channels are VNIR bands, and the others are SWIR bands. After removing the noise bands, the remaining 285 bands are retained. The image consists of 680 × 680 pixels with a spatial resolution of 30 m.

B. Implementation Details

With respect to the CAVE dataset, we select the first 20 HSIs as the training set and the last 12 HSIs as the testing one. Concerning the Harvard dataset, we select the first 30 HSIs as the training set and the rest 20 HSIs as the testing one. HR-HSI and LR-HSI pairs are generated by bicubic downsampling. Meanwhile, due to the limitations of hardware facilities, the original images are cropped, that is, a 32 × 32 patch is randomly cropped for each training, and 16 patches are input into the network as a batch.

To increase the diversity of samples, each patch is randomly flipped horizontally and vertically. To better unify the input of the model, the dataset is normalized between 0 and 1. At the same time, a Gaussian kernel is used to blur the input image, where the kernel size is 5 and the standard deviation is 0.5. Our DCNN network is implemented on the PyTorch framework with NVIDIA A40 GPUs. Specifically, the learning rate is initialized as 1×10^{-4} and is reduced by a factor of 0.1 after every 100 epochs. The training loss is L1 loss. With respect to the hyperparameters of adaptive moment estimation (ADAM) optimization, the exponential decay rates of the first-order β_1 and the second-order β_2 moment estimation are, respectively, set as 0.9 and 0.999, while a small term $\epsilon = 10^{-8}$ is added to the denominator to increase the stability of numerical calculation. The training procedure stops at 200 epochs.

Regarding Chikusei and Yellow River Estuary satellite HSI remote sensing data, the training process of both the sets is consistent with the CAVE one, while the test set is the corresponding SR task. In particular, the left scene in Fig. 6 consists of 2517 × 2335 pixels, in which a local area of 200 × 200 pixels in the lower left corner is selected for testing (marked by the red rectangle), while the rest is taken as the training set. Similarly, the crop YRE dataset center 150 × 150 subarea is considered as test images and the rest is for training. To generate training samples, the LR-HSI data are obtained by applying bicubic downsampling on HR-HSI.

C. Evaluation Metrics

To evaluate the performance of the proposed DCNN approach, we adopt three standard quantitative metrics, i.e., peak SNR (PSNR), structural similarity index (SSIM) [53], and spectral angle mapper (SAM) [54].

D. Ablation Studies

In this section, the ablation experiments are carried out from the following six aspects: the way of combining modules, the convolution type of the model, the number of RFEBs D, the number of ECNBs L, the way of feature fusion strategy, and the choice of loss function. Without loss of generality, a Gaussian filter with random size (three to five) is applied on the training set of the CAVE dataset. The SR scale is fixed as 2.

1) Way of Combining Modules: The proposed DCNN framework consists of four main components, including image reconstruction, shallow feature extraction, feature fusion, and diffused image enhancement. To show the importance of each module in the SR task, various combinations of the four components have been considered, and the results are presented in Table I. Specifically, compared with the complete networks, reconstruction quality degrades when any modules are removed, verifying the necessity of each module and the effectiveness of the proposed DCNN framework. Furthermore, it can be seen from the sixth and last columns in Table I that when HOG and LBP features are introduced and pass through the feature fusion module, PSNR can be increased by 0.899 dB, proving the efficiency of the feature fusion module. Besides, when the diffused image enhancement module is removed, PSNR decreases by 0.36 dB, confirming the indispensability of the diffused component. Therefore, it can be concluded that each module in our DCNN architecture can facilitate the learning process and contribute to the SR task.

2) Convolution Type of the Model: Ablation experiments with respect to the 2-D and 3-D convolutions have been carried out, and the results are provided in Table II. It can be found from Table II that as expected, the performance of 3-D convolution (abbreviated as 3-DC) is better than that of 2-D

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TABLE VII

QUANTITATIVE EVALUATION (AVERAGE PSNR/SSIM/SAM) OF VARIOUS STATE-OF-THE-ART SR ALGORITHMS ON THE CAVE DATASET. THREE SCALE FACTORS ARE CONSIDERED AND BOLD NUMBERS INDICATE THE BEST PERFORMANCE

Scale factor	Evalution metric	Bicubic	GDRRN [51]	SSPSR [52]	MCNet [23]	ERCSR [22]	DCNN (our)
×2	PSNR↑	37.984	38.667	40.733	40.277	40.324	**41.702**
	SSIM↑	0.972	0.977	0.982	0.980	0.980	**0.983**
	SAM↓	3.507	3.541	3.511	3.357	3.361	**3.007**
×4	PSNR↑	32.663	33.941	34.966	34.537	34.649	**35.660**
	SSIM↑	0.922	0.932	0.948	0.943	0.943	**0.949**
	SAM↓	4.941	4.862	4.317	4.210	4.216	**4.200**
×8	PSNR↑	28.680	29.717	30.100	30.765	30.663	**31.857**
	SSIM↑	0.859	0.869	0.883	0.891	0.891	**0.902**
	SAM↓	5.829	5.830	5.875	5.787	5.756	**5.656**

TABLE VIII

QUANTITATIVE EVALUATION (AVERAGE PSNR/SSIM/SAM) OF VARIOUS STATE-OF-THE-ART SR ALGORITHMS ON THE HARVARD DATASET. THREE SCALE FACTORS ARE CONSIDERED AND BOLD NUMBERS INDICATE THE BEST PERFORMANCE

Scale factor	Evalution metric	Bicubic	GDRRN [51]	SSPSR [52]	MCNet [23]	ERCSR [22]	DCNN (our)
×2	PSNR↑	40.538	43.535	44.145	44.618	44.830	**45.162**
	SSIM↑	0.953	0.968	0.969	0.970	0.975	**0.976**
	SAM↓	3.424	3.087	3.045	2.993	2.963	**2.853**
×4	PSNR↑	36.884	38.577	39.043	39.112	39.342	**40.138**
	SSIM↑	0.878	0.901	0.910	0.924	0.928	**0.942**
	SAM↓	3.979	3.696	3.594	3.520	**3.513**	3.518
×8	PSNR↑	33.882	35.411	36.151	36.347	36.388	**36.526**
	SSIM↑	0.874	0.886	0.893	0.896	0.895	**0.911**
	SAM↓	4.335	4.267	4.097	4.064	4.031	**3.997**

convolution (abbreviated as 2-DC). Meanwhile, our developed enhanced convolutional neural (ECN) block has comparable results with 3-DC. It is worth to point out that the number of parameters in ECN is less than one-sixth that in 3-DC, which is valuable for HSI image with large spatial coverage.

3) Number of RFEBs D: The influence of the number of RFEB on the CAVE dataset is listed in Table III. With the increase in number of RFEBs D, the reconstruction quality improves, but the performance declines when $D > 4$. The reason is that the network becomes so deep which is difficult to train. Considering the trade-offbetween performance and complexity, we set the number of RFEBs D to be 4 in the subsequent experiments.

4) Number of ECNBs L: Table IV analyzes the influence of the number of ECNB L on the CAVE dataset. Three evaluation metrics are used, and the scale factor is 2. It can be observed from Table IV that when the number of ECNBs L is 2, the results are the worst, which is mainly due to that the network level is not deep enough, and the structural information cannot be fully learned. When $L > 2$, the recovery effect of the network is basically the same. Therefore, according to the tradeoff between the recovery performance and the amount of parameters, the number of ECNBs L is set as 3 in subsequent experiments.

5) Way of Feature Fusion Strategy: The combination manner impact of the three kinds of features is concerned, and the results are listed in Table V. "Addition" and "concatenation," respectively, denote the simple addition and concatenation fusion of the three kinds of features. It is worth to point out that the concatenation strategy corresponds to the same weight manner. Considering our DCNN method, instead of directly concatenating all the three kinds of features, both HOG and LBP are combined together, and then a number of convolution operations are applied. After that, the achieved features are combined with the deep features. It can be found from Table V that our fusion strategy is better than the rest two manners.

6) Choice of Loss Function: In the supervised HSI SR task, the goal is to make the SR image as close as possible to the real hyperspectral high-resolution image. $L2$ loss is calculated as the sum of the squares of the absolute difference between the actual value and the target value, while $L1$ loss is calculated as the sum of the absolute difference between the actual value and the target value. Table VI lists the influence of different loss functions on the model, and a combination of both the loss functions is also concerned. It can be clearly seen from Table VI that $L1$ loss outperforms the other two loss functions.

E. Results on CAVE and Harvard Dataset

In this section, we adopt two public HSI datasets (CAVE and Harvard) to compare the performance of DCNN with a number of alternatives, including bicubic [55], grouped deep recursive residual network (GDRRN) [51], spatial-spectral prior network based super-resolution network (SSPSR) [52], mixed convolutional network (MCNet) [23], and

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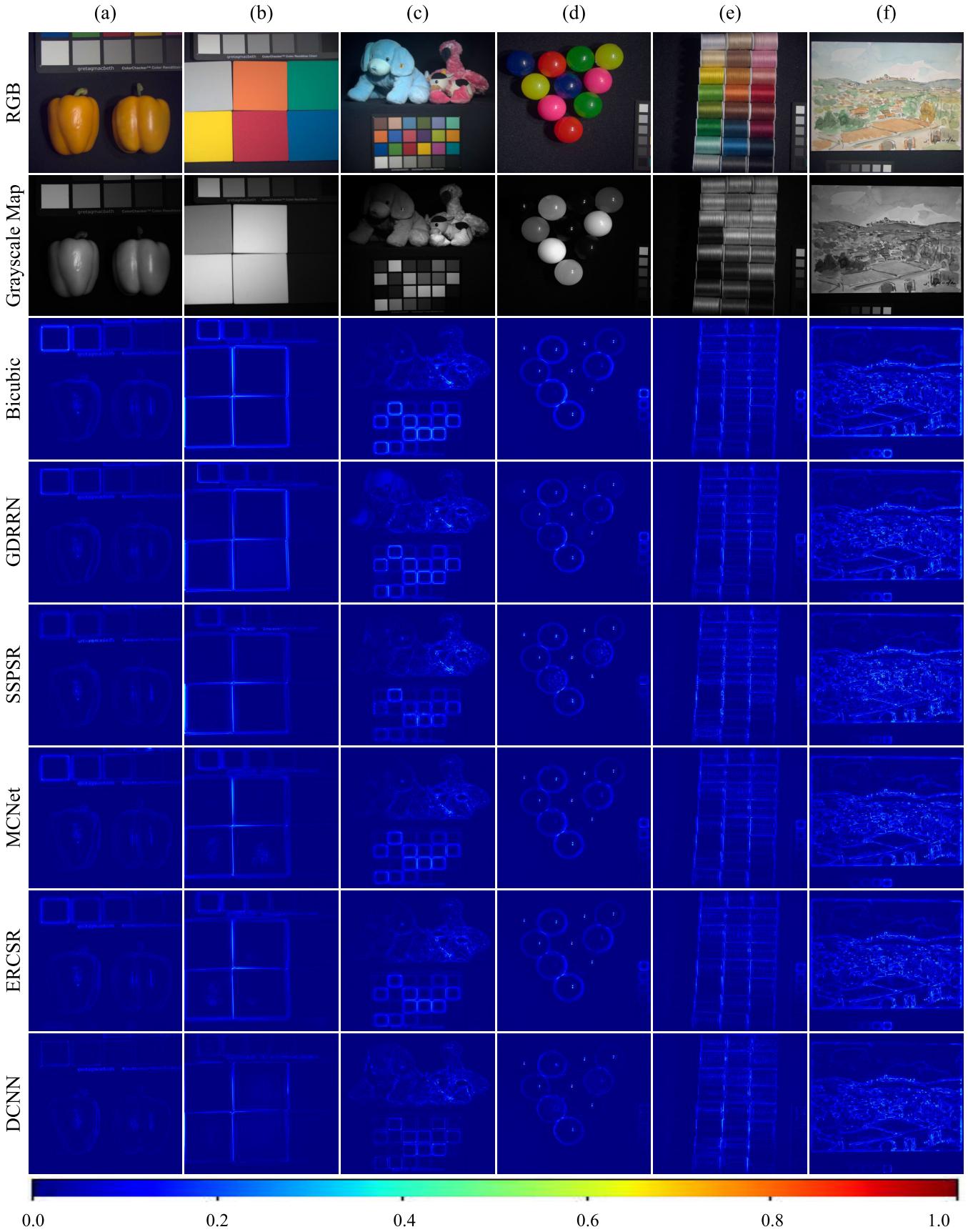


Fig. 7. Error map visualization of six compared methods on the CAVE dataset (the scale factor is 8), where band images at 640, 660, 680, 620, 690, and 590 nm are selected from left to right for display. (a) real and fake peppers, (b) sponges, (c) stuffed toys, (d) superballs, (e) thread spools, and (f) watercolors.

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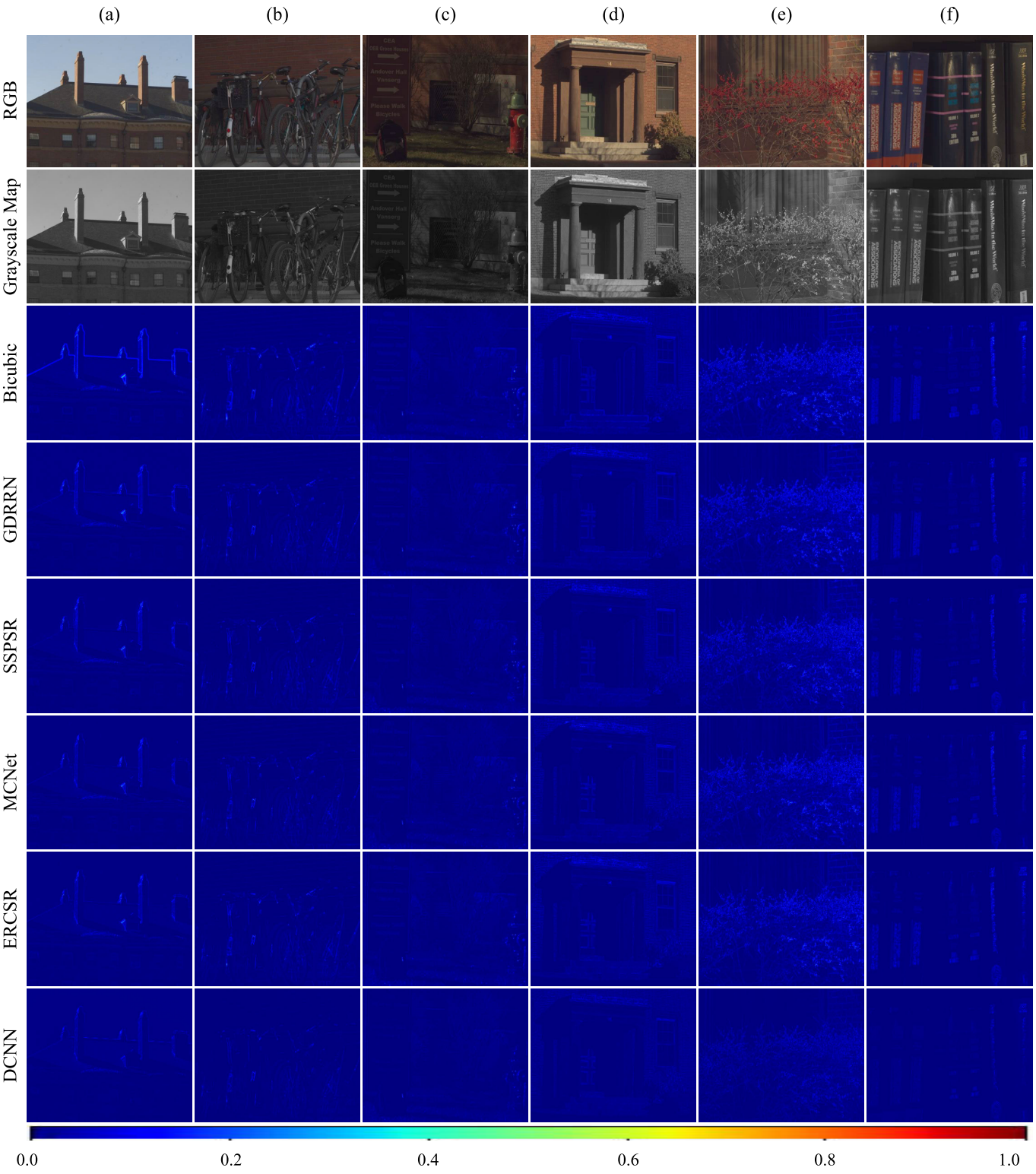


Fig. 8. Error map visualization of six compared methods on the Harvard dataset (the scale factor is 8), where band images at 620, 490, 580, 510, 630, and 530 nm are selected from left to right for display. (a) image f4, (b) image f5, (c) image f6, (d) image f7, (e) image f8, and (f) image h2.

exploring the relationship between 2D/3D convolution (ERCSR) [22]. Quantitative evaluation (average PSNR/SSIM/SAM) of various state-of-the-art SR algorithms on the CAVE and Harvard datasets is displayed in Tables VII and VIII, respectively. Three scale factors are considered and bold numbers indicate the best performance. Moreover, parameter number of different models on the CAVE dataset is listed in Table IX.

With respect to the CAVE dataset, it can be found from Table VII that bicubic and GDRRN provide the worst in these methods. Due to the simple structure of the two algorithms, it is difficult to learn sufficient spatial and spectral information

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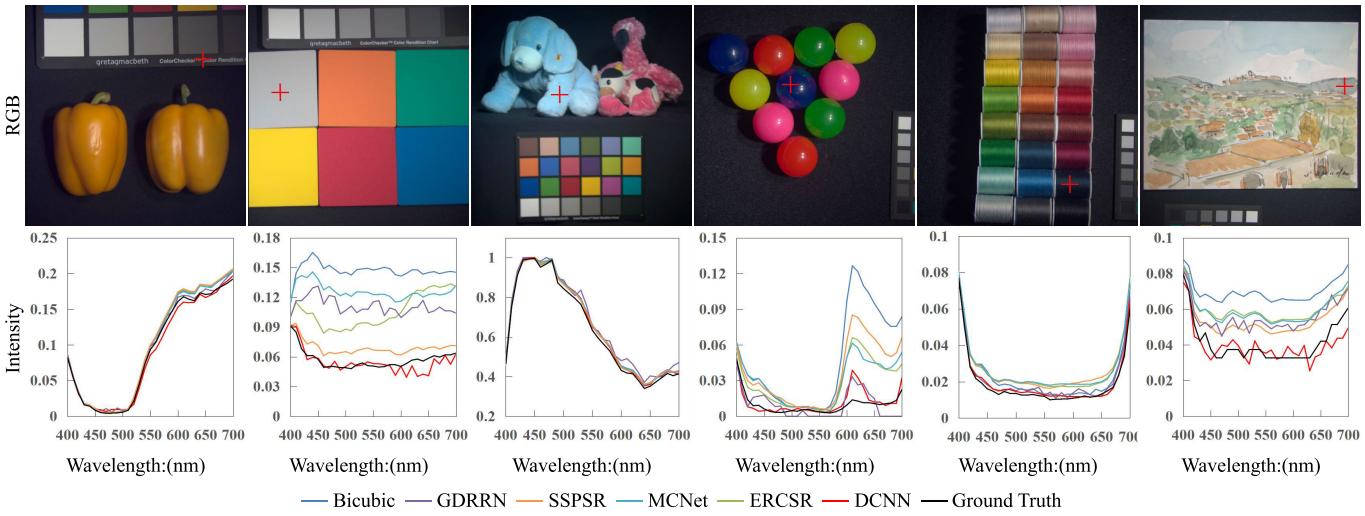


Fig. 9. Visualization of the spectral restoration on the CAVE dataset, where selected pixel locations are marked with red crosses in the RGB image.

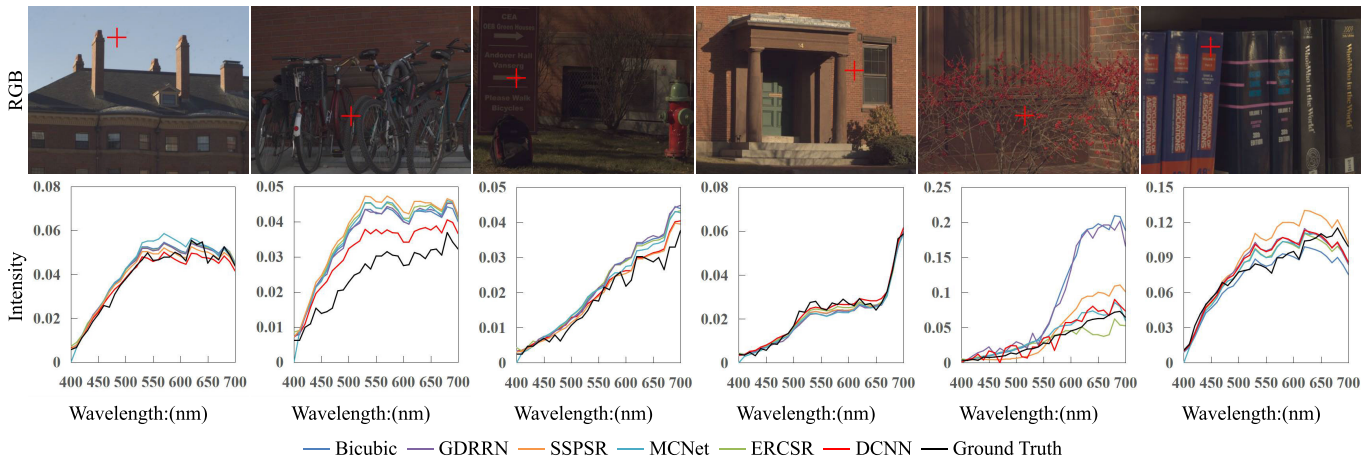


Fig. 10. Visualization of the spectral restoration on the Harvard dataset, where selected pixel locations are marked with red crosses in the RGB image.

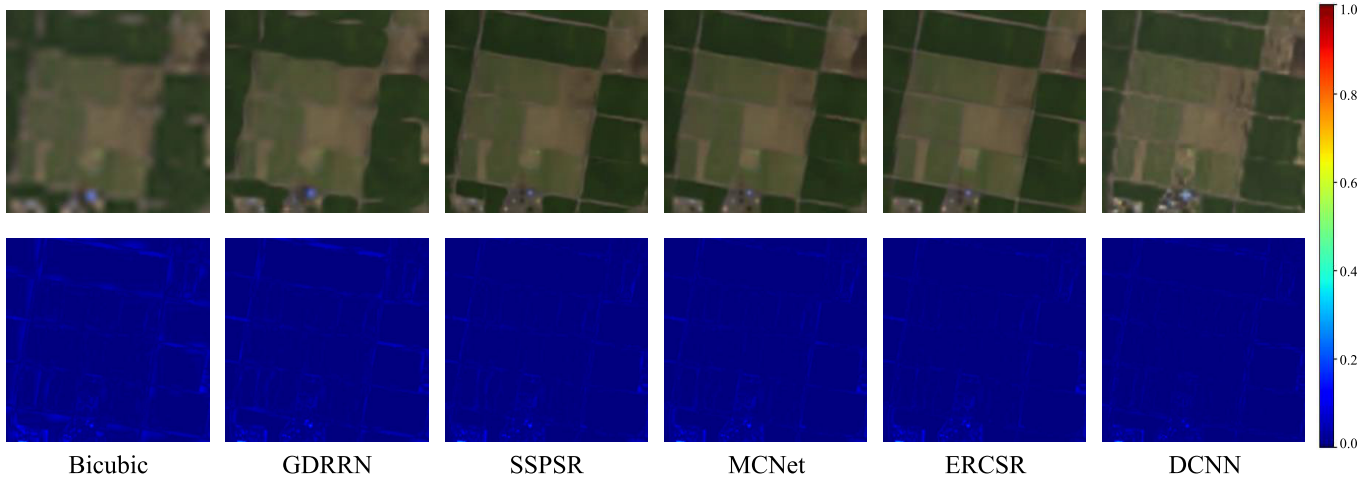


Fig. 11. With an upsampling factor of 8, the false color images in the top row are reconstructed by various compared methods (here bands 50-38-19 are used as R-G-B channels). The second row displays the corresponding error maps.

of HSI. Second, the PSNR of SSPSR on $\times 2$ and $\times 4$ is better than the above two methods in Table VII, even higher than MCNet and ERCSR that are based on 3-D convolution, which is attributed to the extremely large number of SSPSR parameters, shown in Table IX. Moreover, the proposed DCNN approach strengthens the learning of joint spatial–spectral information through each module and significantly outperforms other state-of-the-art methods in three scale factors.

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TABLE IX
PARAMETER NUMBER OF DIFFERENT MODELS ON THE CAVE DATASET

method	params (10^5)
GDRRN	3.66
SSPSR (r = 2)	105.34
SSPSR (r = 4)	128.94
SSPSR (r = 8)	152.54
ERCSR (r = 2)	13.49
ERCSR (r = 4)	15.95
ERCSR (r = 8)	23.81
MCNet (r = 2)	19.28
MCNet (r = 4)	21.74
MCNet (r = 8)	29.61
DCNN (r = 2)	12.41
DCNN (r = 4)	13.88
DCNN (r = 8)	15.36

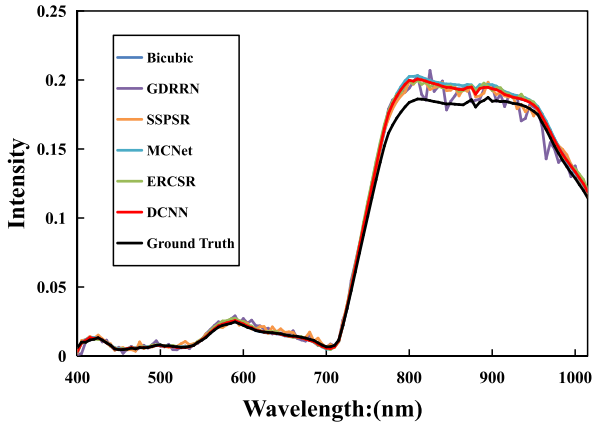


Fig. 12. Spectral restoration of various compared methods on the Chikusei dataset. The spectral curve corresponds to spatial position (134, 1678).

Similarly, concerning the results on the Harvard dataset listed in Table VIII, bicubic and GDRRN still achieve the worst results. However, all the indicators for MCNet and ERCSR in the Harvard dataset are higher than those for SSPSR. This is because the Harvard dataset has more samples and contains more texture information compared with the CAVE dataset. In particular, 3-D convolution can effectively learn more texture and spectral information, so as to have a higher reconstruction effect. Likewise, our DCNN method outperforms all the competitors in most cases, verifying its superiority.

With respect to the CAVE and Harvard HSI datasets, the error maps of comparative approaches, which reflect the difference between estimated images and the reference ones, have been visually displayed in Figs. 7 and 8 (the scale factor is 8), respectively. The smaller the error map, the higher performance the SR method. Specifically, the proposed DCNN approach has a good recovery of texture detailed areas, such as the doll in Fig. 7(c) and the flower in Fig. 8(e). Furthermore, DCNN performs well on smooth textured regions, such as bell peppers in Fig. 7(a) and book surfaces in Fig. 8(f), and it has also achieved good results in the recovery of edge area, such as the color card in Fig. 7(c) and the roof in Fig. 8(a). Compared with other state-of-the-art methods, DCNN produces better results in most cases. This is mainly attributed to the fact that DCNN can retain rich spatial information through HOG and LBP feature descriptors and exploit the global similarity by the diffused image enhancement module.

We also visualize the spectral curves of reconstructed images from the CAVE and Harvard datasets with a scale factor of 8. As illustrated in Figs. 9 and 10, the pixel locations are randomly selected (the corresponding pixel locations are marked with red cross in the RGB image). It can be clearly seen from Figs. 9 and 10 that the spectral curves reconstructed by our DCNN are closer to the ground truth than the compared ones in most cases. In particular, the robustness of the proposed DCNN on spectral restoration aspect is in favor of the developed multilayer RFEB. In the proposed DCNN approach, a number of RFEB modules are concatenated together and a 1×1 convolution is adopted in the skip connection, which greatly boosts the spectral fidelity.

F. Results on Remote Sensing Dataset

1) Chikusei Dataset: Quantitative results of various state-of-the-art SR algorithms on the Chikusei dataset are listed in Table X. It can be easily found that the two 2-D CNN-based methods (SSPSR and the proposed DCNN) have shown superior performance than the two 3-D CNN-based ones (MCNet and ERCSR). This is mainly due to that both MCNet and ERCSR reduce the amount of parameters by a dimensionality reduction step, which inevitably deteriorates the spectral information. Meanwhile, the discontinuity of spectral bands (caused by the omission of water absorption and noise interference) also degrades the capability of 3-D CNN-based methods. Fortunately, by adopting the 2-D convolution operator, SSPSR and our DCNN can make full use of the spectral representation, and thus the spectral features can be well-restored. Furthermore, our DCNN approach outperforms other state-of-the-art methods on all three scale factors.

Fig. 11 displays the reconstructed images and error maps by the compared methods on the Chikusei dataset, where the scale factor is 8. It can be observed from Fig. 11 that both bicubic and GDRRN methods produce a lot of artifacts and noises, since these two methods are too simple in design and cannot extract features well. Similar conclusion can be made that the proposed DCNN approach has the least errors among the compared methods. Besides, spectral restoration of various compared methods on the Chikusei dataset is visualized in Fig. 12, where the spectral curve corresponds to spatial position (134, 1678). It can be easily found that the spectral curve restored by the proposed DCNN method is closer to the ground truth, which is also very stable. This is mainly due to the introduced diffused image enhancement module and the multi-layer RFEB, which greatly boosts the spectral fidelity.

2) Yellow River Estuary Dataset: In this section, an HSI remote sensing image acquired by China's Gaofen5 satellite is used to demonstrate the effectiveness of the proposed DCNN approach. As displayed in Fig. 13, the false color image

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TABLE X
QUANTITATIVE EVALUATION (AVERAGE PSNR/SSIM/SAM) OF VARIOUS STATE-OF-THE-ART SR ALGORITHMS ON THE CHIKUSEI DATASET. THREE SCALE FACTORS ARE CONSIDERED AND BOLD NUMBERS INDICATE THE BEST PERFORMANCE

Scale factor	Evalution metric	Bicubic	GDRRN [51]	SSPSR [52]	MCNet [23]	ERCSR [22]	DCNN (our)
×2	PSNR↑	45.152	49.429	50.709	50.160	50.061	**51.132**
	SSIM↑	0.986	0.994	**0.996**	**0.996**	**0.996**	**0.996**
	SAM↓	1.231	0.886	0.717	0.753	0.754	**0.693**
×4	PSNR↑	38.292	42.396	42.552	42.359	42.559	**42.751**
	SSIM↑	0.947	0.975	0.982	0.982	0.983	**0.986**
	SAM↓	2.475	1.615	1.320	1.415	1.340	**1.301**
×8	PSNR↑	34.557	36.539	38.136	37.731	38.041	**38.375**
	SSIM↑	0.906	0.933	0.959	0.957	0.959	**0.963**
	SAM↓	3.952	3.005	2.231	2.486	2.258	**2.176**

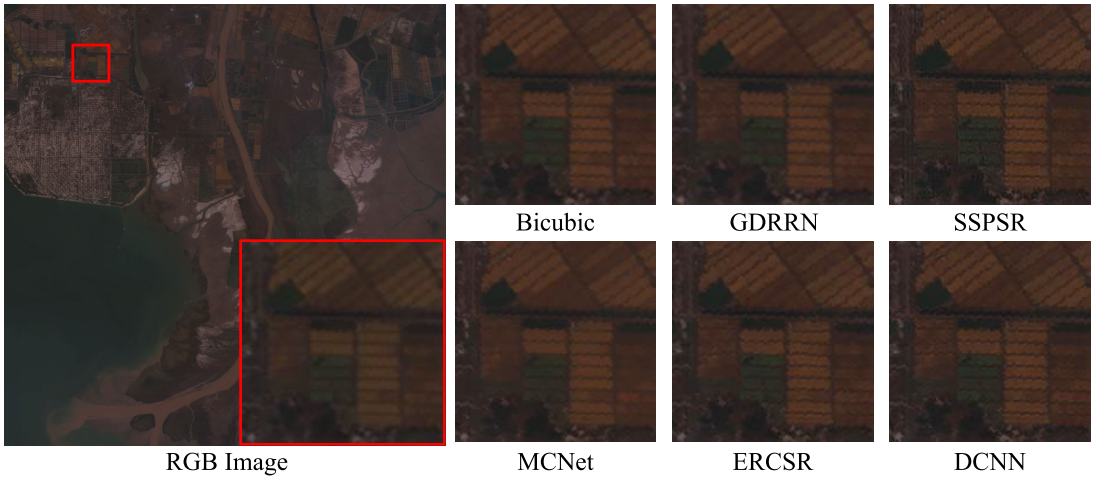


Fig. 13. With an upsampling factor of 2, the false color image is reconstructed with bands 65-30-3 as R-G-B channels.

corresponds to the Yellow River Estuary area, which is reconstructed with bands 65-30-3 as R-G-B channels. The spatial size is 680 × 680, which contains 285 spectral bands. Due to hardware limitations, we perform SR on the original image for scale factor ×2, and the results of compared methods are evaluated by visual effects owing to the lack of ground truth. The red rectangle part contains rich texture information and is a challenging issue for image SR, and the SR results of various techniques are displayed in Fig. 13. Concretely, it can be observed from Fig. 13 that GDRRN and MCNet provide oversmoothed contours on the restoration field, while SSPSR and ERCSR cause image distortion in the SR results. Overall, our DCNN approach can improve the clarity of the image and has strong adaptability.

V. CONCLUSION

In this article, we have proposed a DCNN for HSI SR. First, the developed diffused image enhancement module is able to sufficiently exploit the global similarity of image, which fine-tunes the reconstructed image by making use of the overall information, driving the SR image closer to the real HSI, and effectively suppressing noise and artifacts. Second, a multilayer RFEB has been carefully designed. A series of spectral convolutions are adopted to learn spectral features at different depths, which has a good performance in spectral restoration. Meanwhile, a number of RFEB modules are concatenated together and a 1 × 1 convolution is adopted in the skip connection, which greatly boosts the spectral fidelity, while network complexity and memory burden are reduced as well.

Third, through incorporating both kinds of shallow feature descriptors (HOG and LBP are used to retain the shape and texture information of image, respectively), the spatial structure of objects can be well-represented, which plays an important role in SR reconstruction. Furthermore, to effectively make use of the extracted shallow and deep features, a feature fusion strategy is used to reinforce the reconstruction efficiency. Extensive experiments are conducted on a wide range of HSI datasets, including two natural scenes and two remote sensing ones, and a number of state-of-the-art methods are considered in comparison. The experimental results constantly demonstrate that the proposed DCNN method achieves the best SR performance in most cases. Our DCNN not only has a good effect on the shape representation of objects but also restores the internal texture information.

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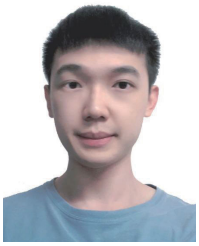
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